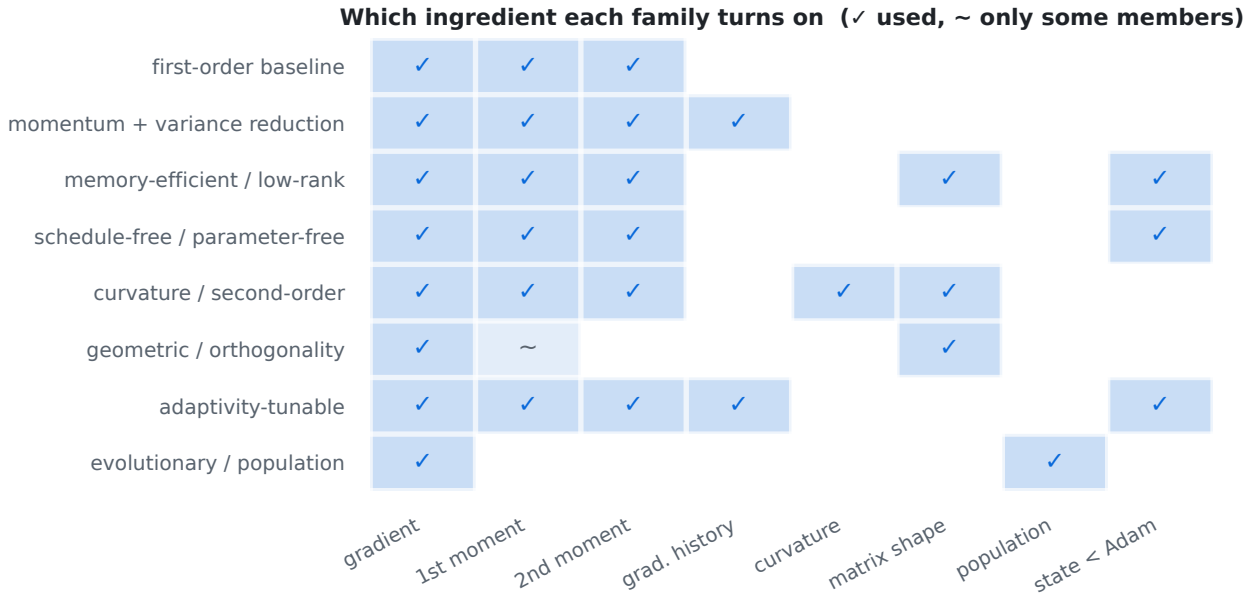
Which ingredient each family turns on (✓ used, ~ only some members)
first-order baseline
momentum + variance reduction
memory-efficient / low-rank
schedule-free / parameter-free
curvature / second-order
geometric / orthogonality
adaptivity-tunable
evolutionary / population
gradient
1st moment
2nd moment
grad. history
curvature
matrix shape
population
state < Adam